



Photometric Image Formation and Digital Camera

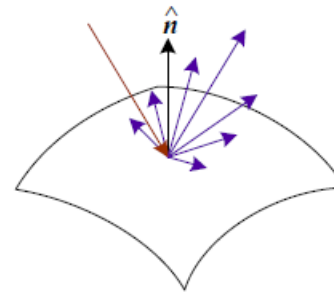
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Outline

- **Photometric image formation**

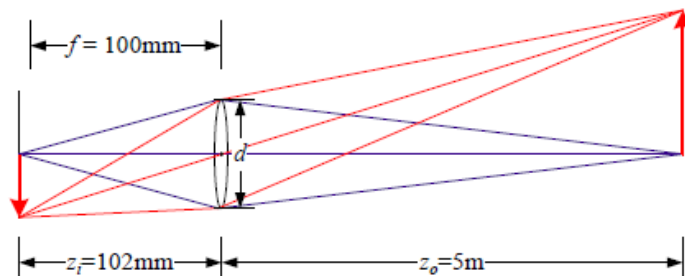
- Chap 2.2



(1st edition)

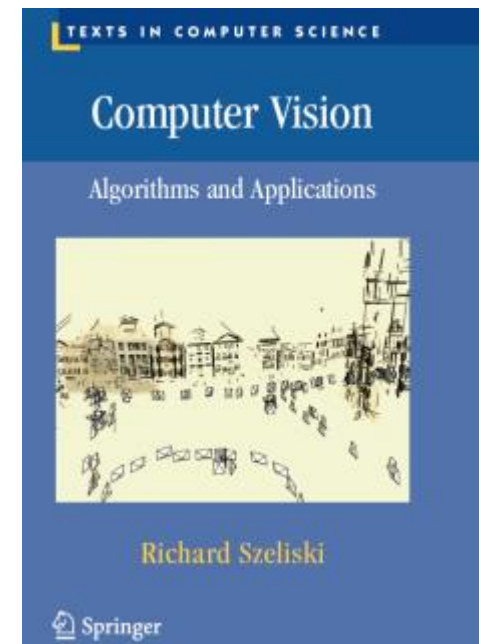
- **Digital camera**

- Chap 2.3



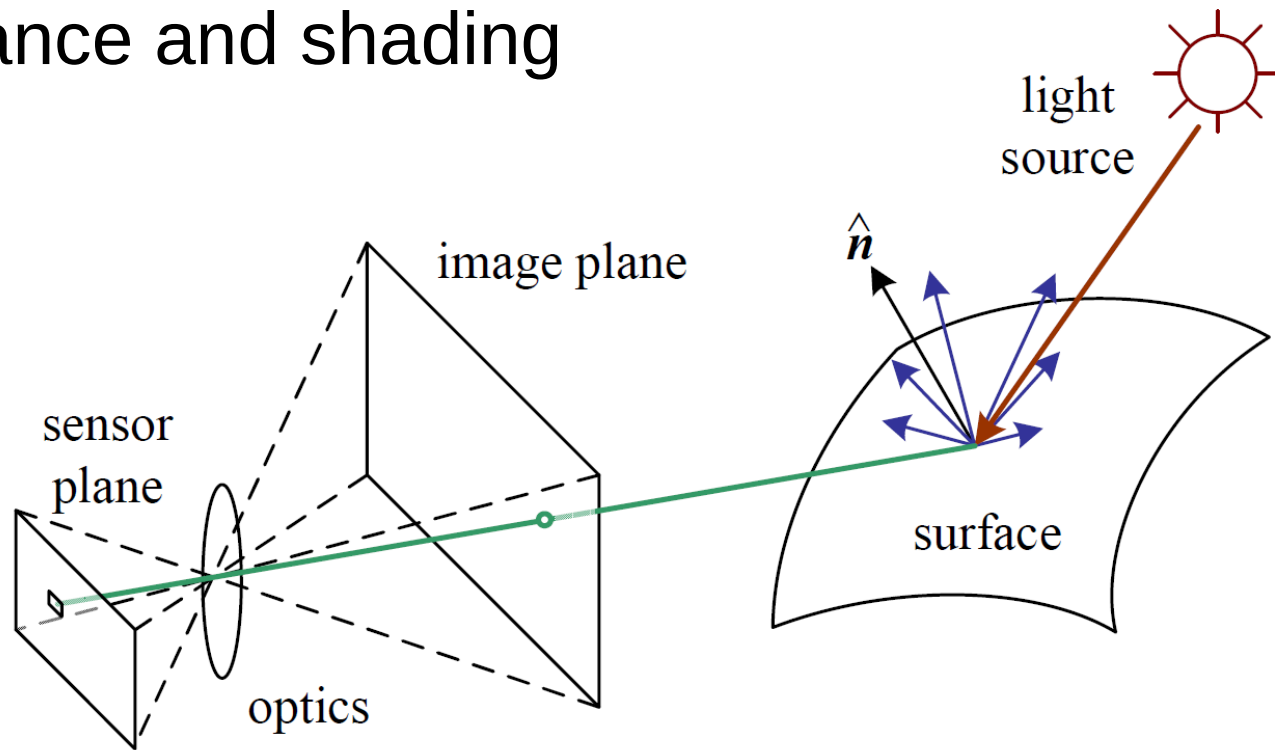
G	R	G	R
B	G	B	G
G	R	G	R
B	G	B	G

Scientific models for camera



Photometric image formation

- Lighting $L(\lambda)$ Radiance $L(\hat{\mathbf{v}}; \lambda)$ Directional
- Reflectance and shading
- **Optics**

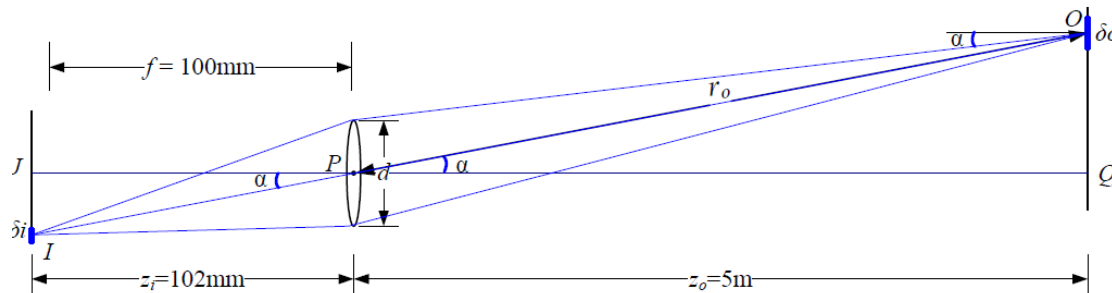


Optics

- (Thin) Lens law
- F-number (N)
 - Aperture diameter $d = f/N$
- One full stop of f-number $\Rightarrow$ twice pupil area
 - $f/1.4, f/2, f/2.8, \dots (\sqrt{2}x)$
 - One full stop $\Rightarrow$ Half of shutter speed
 - One full stop $\Rightarrow$ one incremental EV (exposure value)

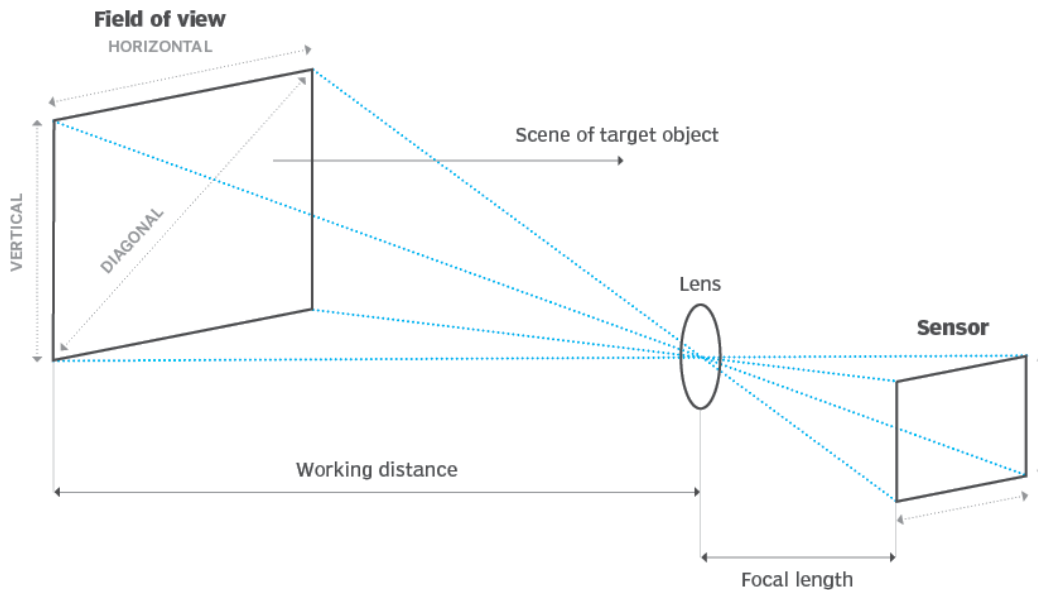
$$\frac{1}{z_o} + \frac{1}{z_i} = \frac{1}{f}$$

$$f/\# = N = \frac{f}{d}$$

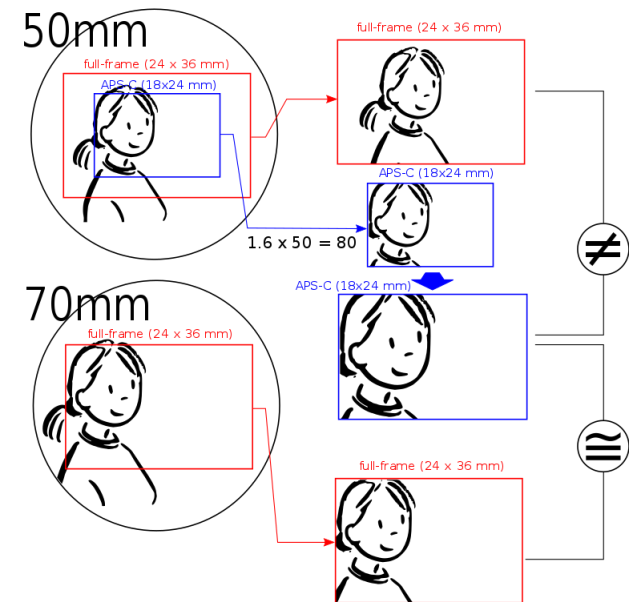


Field-of-view (FOV)

- Captured area (in angle) by camera
 - Wide angle $\Leftrightarrow$ short f (higher input radiance)
 - Telephoto $\Leftrightarrow$ long f (lower input radiance)



Ref: TechTarget.

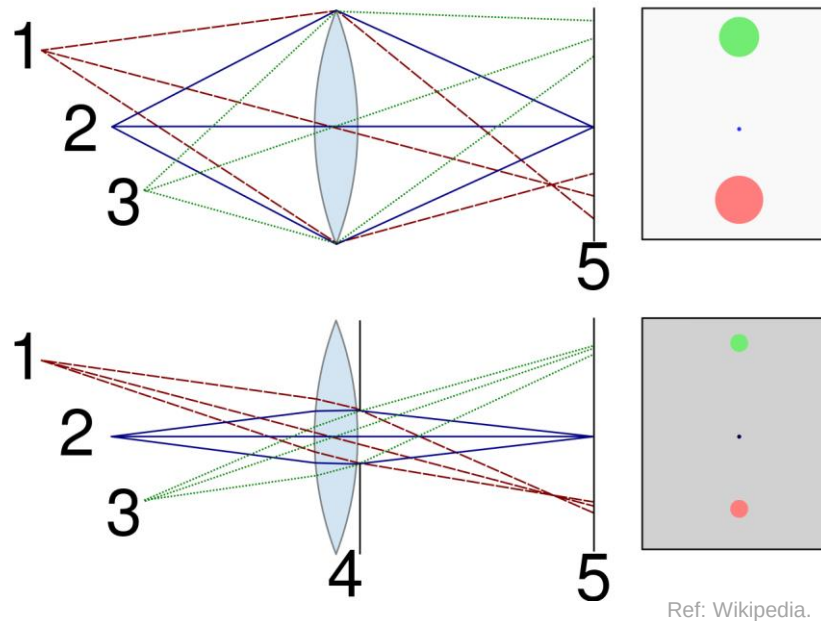


Ref: Wikipedia.

Depth-of-field (DoF)

- Depth-of-view

- Distance in between where objects are in sharp focus
- Long f /large aperture $\Rightarrow$ shallow DoF
- Short f /small aperture $\Rightarrow$ deep DoF



Aperture = $f/1.4$. DOF=0.8 cm



Aperture = $f/22$. DOF=12.4 cm

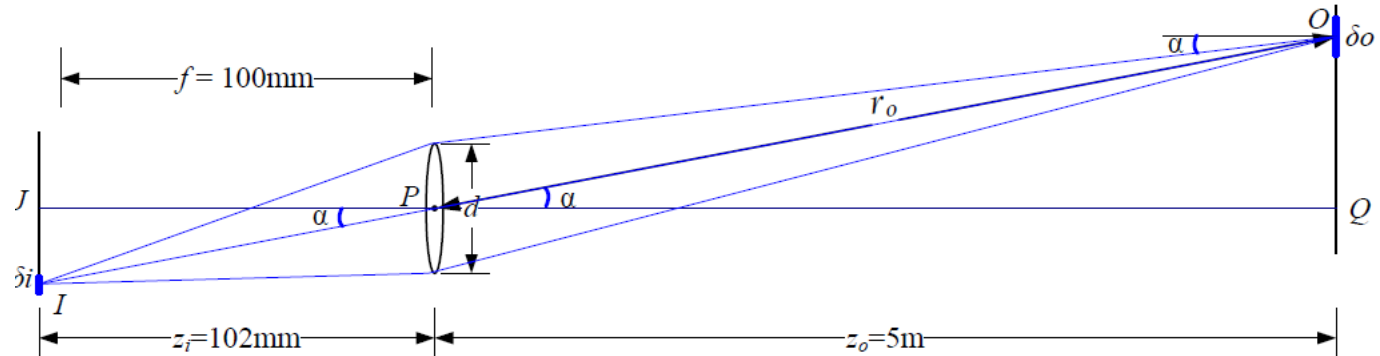
Aberrations

- Real lens not infinitely thin
 - Seidel aberrations from third-order optics
- Refraction index varies with wavelength
 - Chromatic aberration (varied focal length)
- Compound lenses are used to mitigate the above aberrations
 - The single-lens model is only used for analysis



Vignetting

- Natural



$$E = L \frac{\pi}{4} \left(\frac{d}{f} \right)^2 \cos^4 \alpha$$

E: irradiance

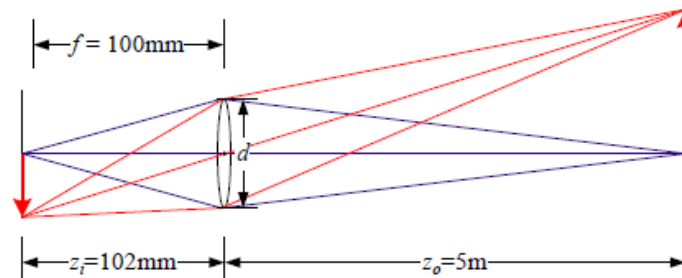
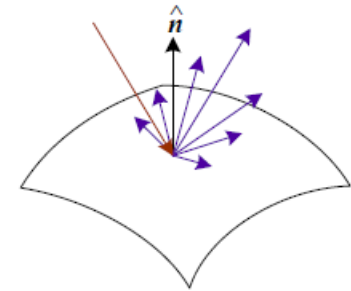
L: radiance



- Mechanical
 - Internal occlusion of camera (lens)

Outline

- Photometric image formation
 - Chap 2.2
- **Digital camera**
 - Chap 2.3

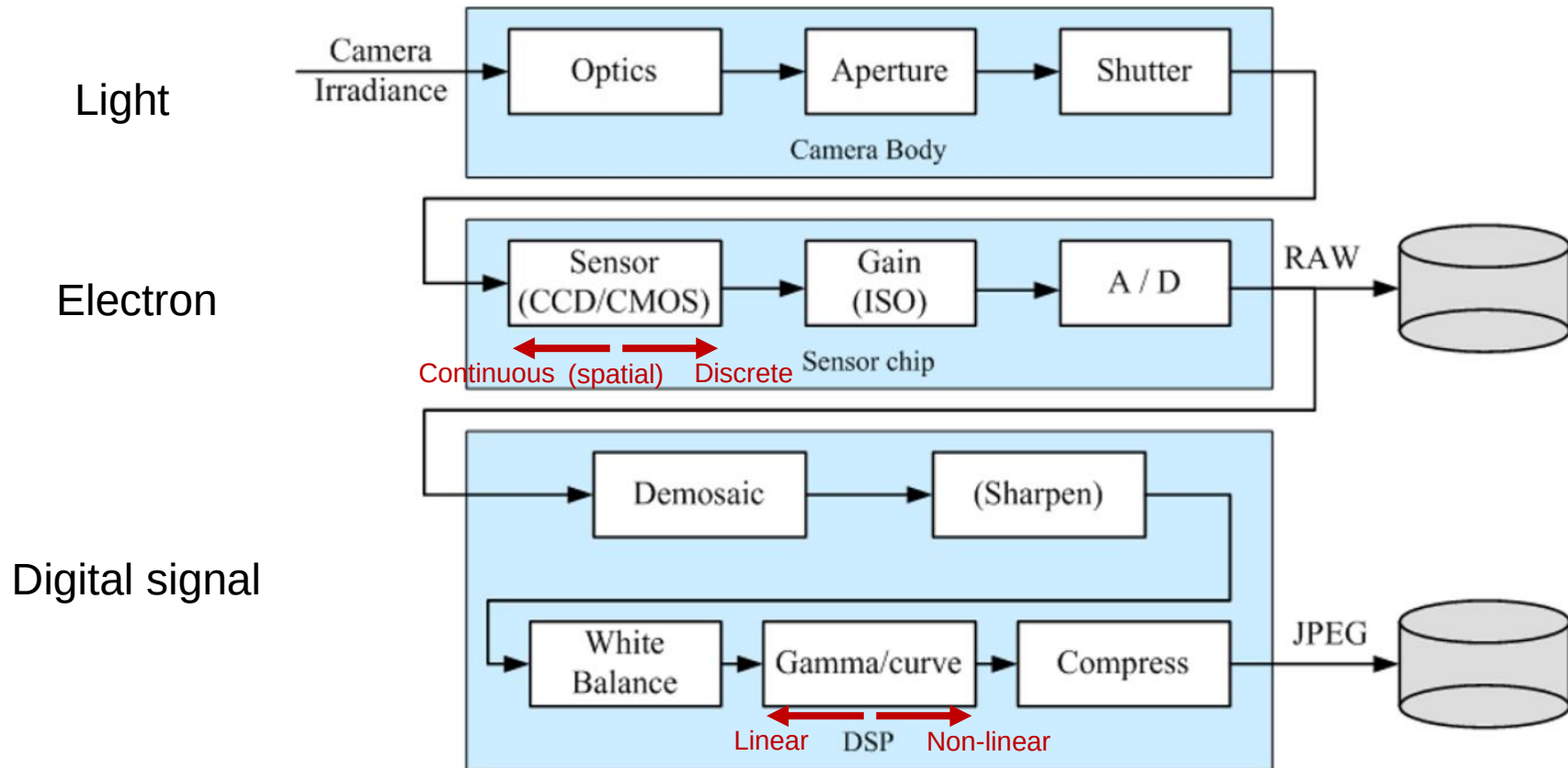


G	R	G	R
B	G	B	G
G	R	G	R
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Scientific models for camera

Digital Camera

- Complex scientific model in three parts



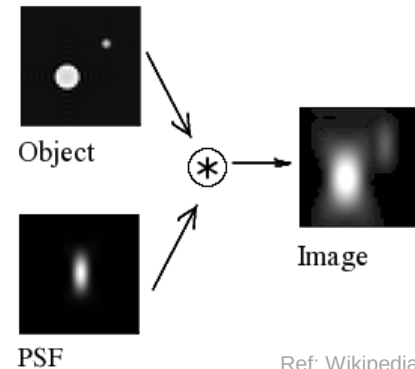


Sensor

- Type
 - CCD: passive-pixel sensor, higher fill factor
 - CMOS: active-pixel sensor, lower fill factor
 - Could be changed using 3D IC technology
- Fill factor
 - Active sensing area over available area
 - Higher is preferable (more light and less aliasing)
- Analog gain
 - ISO setting: 100, 200, 400, ... (confusing)
- Noise
 - Shot noise, amplifier noise, quantization noise (hard)
 - Fixed pattern noise, dark current noise (easy)

Sampling and Aliasing

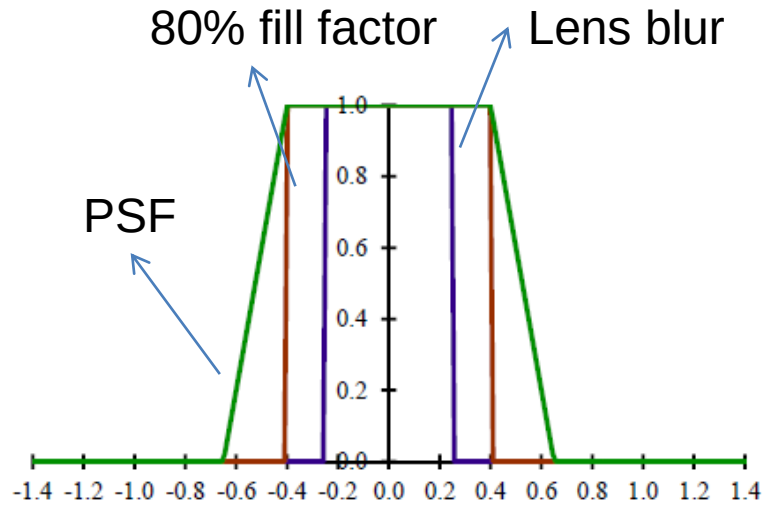
- Point spread function (PSF)
 - Response of an ideal point source
 - Consists of
 - (Static) Convolution of lens blur and finite active pixel area
 - (Limit) Airy disk due to diffraction
 - (Dynamic) camera shake
- Modulation transfer function (MTF)
 - Magnitude of FT of PSF
- Image pixels have aliasing inevitably due to fill factor $\leq 100\% < 400\%$



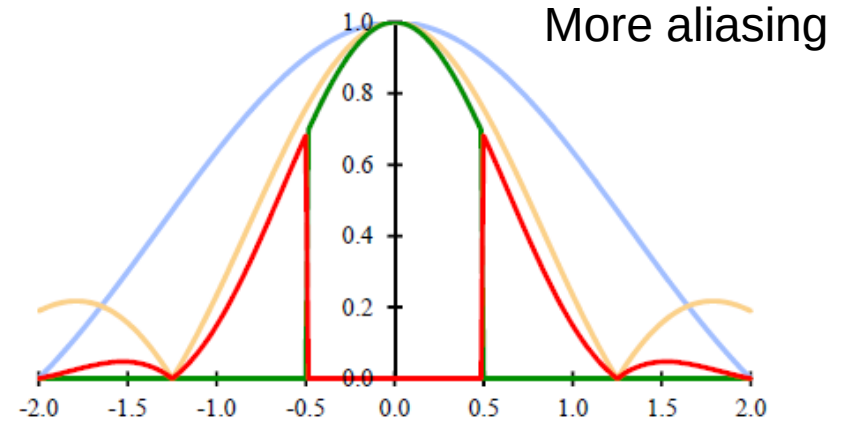
Ref: Wikipedia.

$$f_s \geq 2f_{\max}$$

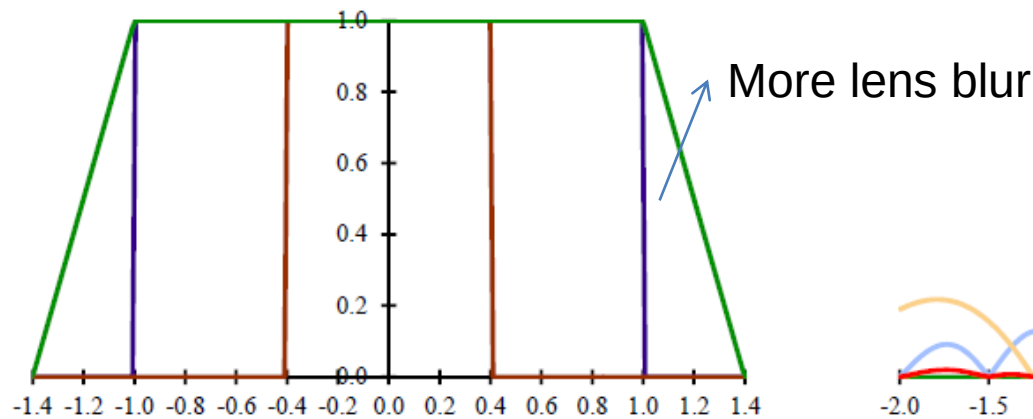
Aliasing Examples (Trade-off)



(a)

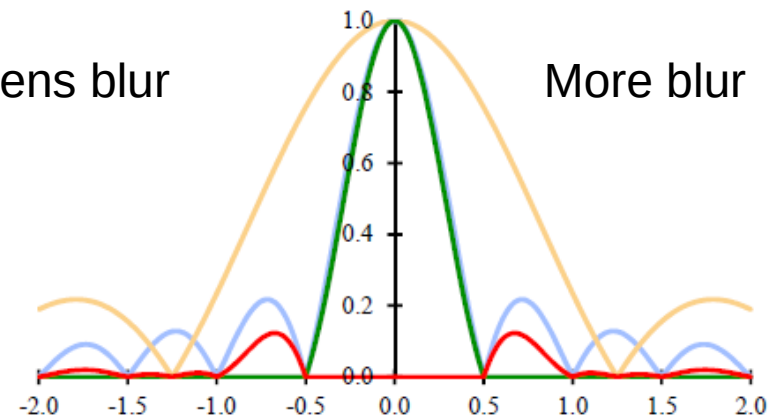


(b)



(c)

PSF

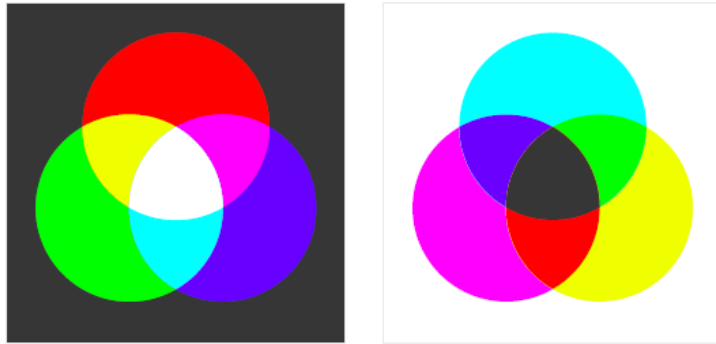


(d)

MTF

Color

- **Additive vs. Subtractive colors**



- Linear domain
 - CIE RGB, XYZ, Color transformation, color filter array
- Non-linear domain
 - Gamma, $L^*a^*b^*$, $Y'CbCr$, ...

“To be linear, or not to be?”
is always the question.

Color Filter Arrays

G	R	G	R
B	G	B	G
G	R	G	R
B	G	B	G

Bayer Pattern
(RAW format)

rGb	Rgb	rGb	Rgb
rgB	rGb	rgB	rGb
rGb	Rgb	rGb	Rgb
rgB	rGb	rgB	rGb

Demosaiced RGB

Color Balance

- What color will you get when yellow light on white objects (e.g. book pages)?
 - But we humans perceive it as white, and demand the picture also look white
 - Usually achieved by per-component scaling (not a scientific model)





Gamma

- Non-linear mapping before discretization
- Origin
 - CRT phosphors have non-linear response between input voltage and brightness

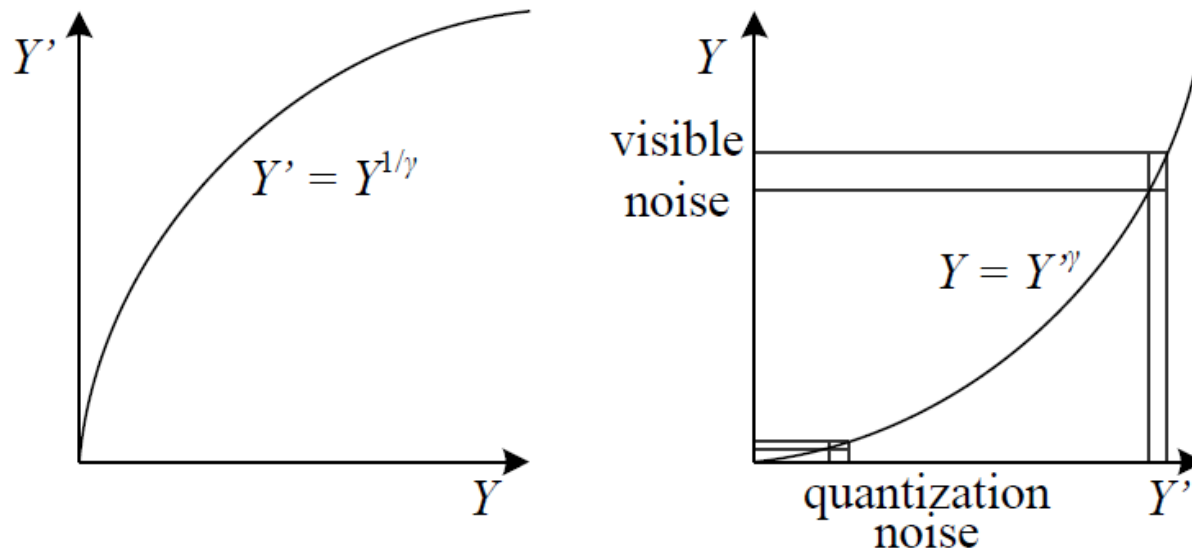
$$B = V^\gamma \quad \gamma = 2.2 \sim 2.4$$

- So we would pre-map the luminance (gamma correction)

$$Y' = Y^{\frac{1}{\gamma}}$$

Gamma

- Why we still use gamma correction now?
- Its map suitable for human perception sensitivity





L*a*b* color space

- (nearly) perceptually uniform
 - Human perceives relative luminance difference of about 1%

$$L^* = 116f\left(\frac{Y}{Y_n}\right) \quad \text{lightness} \quad \begin{aligned} a^* &= 500 \left[f\left(\frac{X}{X_n}\right) - f\left(\frac{Y}{Y_n}\right) \right] \\ b^* &= 200 \left[f\left(\frac{Y}{Y_n}\right) - f\left(\frac{Z}{Z_n}\right) \right] \end{aligned}$$

(X_n, Y_n, Z_n) is the measured white point

$$f(t) = \begin{cases} t^{1/3} & t > \delta^3 \\ t/(3\delta^2) + 2\delta/3 & \text{else,} \end{cases}$$

Non-linear transform
(like gamma?)

Linear vs. Non-linear Coding

(coded in linear luminance)

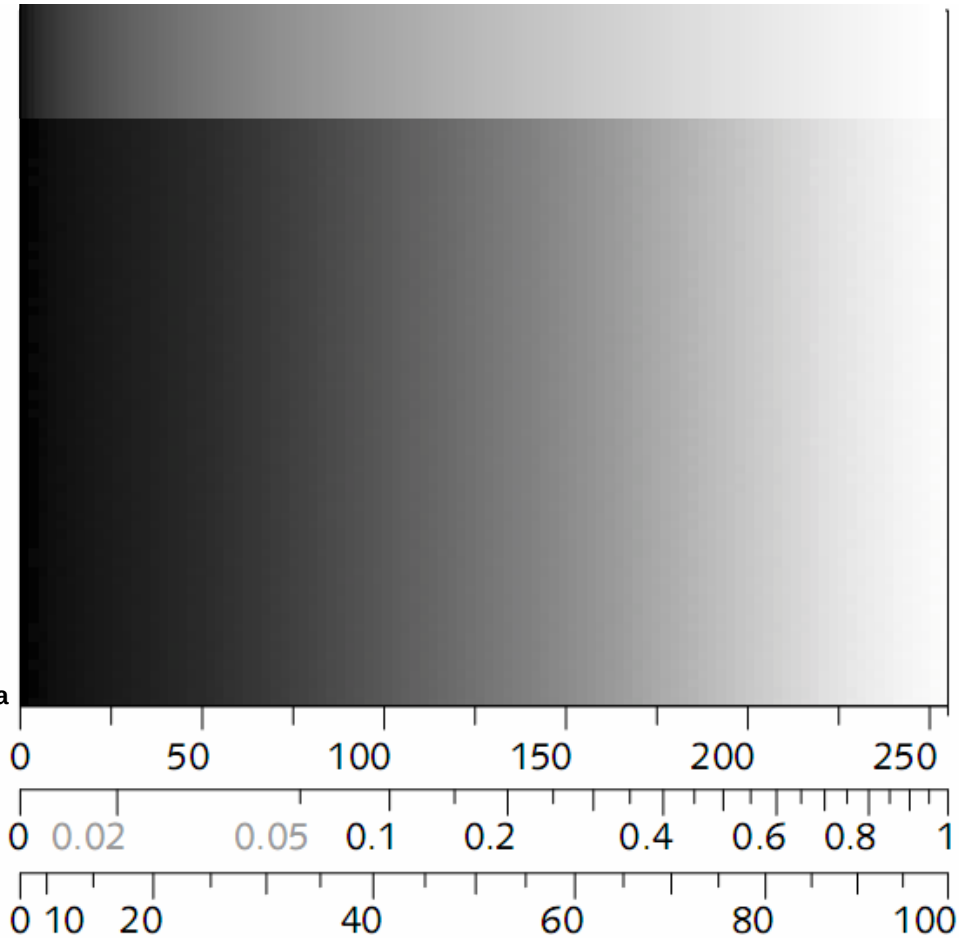
(Gamma 2.5 corrected coding)

Pixel Value $\propto$ Luminance^{1/gamma}

Pixel value, 8-bit scale

Luminance, relative

CIE Lightness, L^*



Ref: Digital Video and HDTV



Y'CbCr color space

- Widely used in image/video storage
- Rec. 601 luma/chroma matrix

$$\begin{bmatrix} Y' \\ C_b \\ C_r \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.168736 & -0.331264 & 0.5 \\ 0.5 & -0.418688 & -0.081312 \end{bmatrix} \begin{bmatrix} R' \\ G' \\ B' \end{bmatrix} + \begin{bmatrix} 0 \\ 128 \\ 128 \end{bmatrix}$$

↑
RGB after
gamma correction
(the prime ' indicates non-linearity)



Compression

- Lossless
 - Camera RAW, PNG, BMP
- Lossy
 - Usually in Y'CbCr and use block-based coding
 - Image: JPEG
 - Video: MPEG-1/2/4, H.264, WMV, H.265



Appendix

Radiance L

- The “light ray”
 - Note the definition with the cross-sectional area leads to the $\cos\theta$ term in many equations

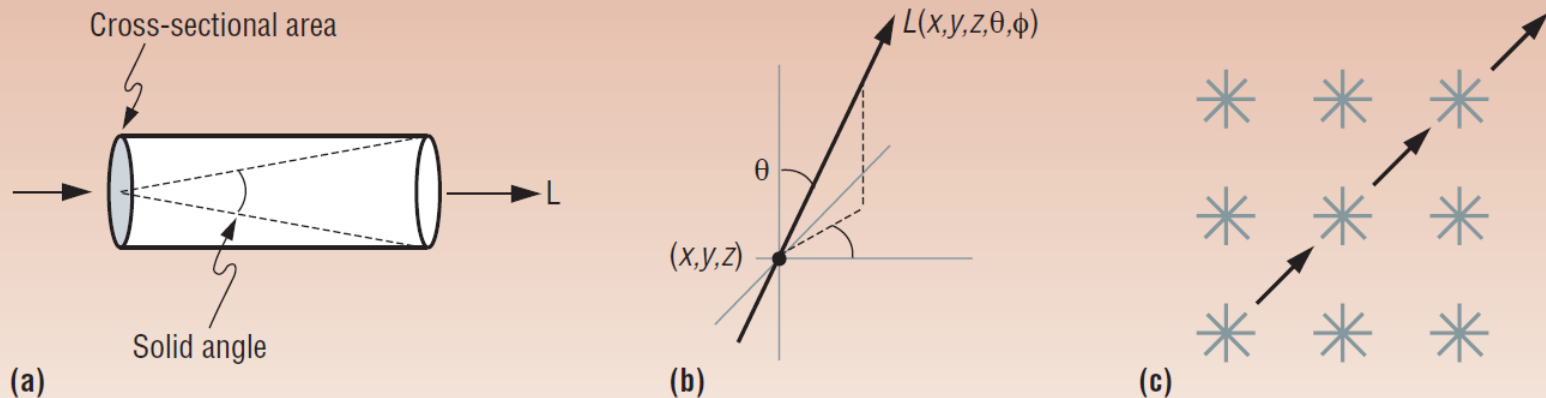


Figure 1. The 5D plenoptic function, representing the flow of light through 3D space. (a) Radiance L along a ray can be thought of as the amount of light traveling along all possible straight lines through a tube whose size is determined by its solid angle and cross-sectional area. (b) Parameterizing a ray by position (x, y, z) and direction (θ, ϕ) . (c) Radiance along a ray remains constant if there are no blockers. This leads to redundancy in the plenoptic function.

Ref : M. Levoy, “Light fields and computational imaging,” *IEEE Computer*, 2006.



Digital camera 3A

- AE: automatic exposure
 - F-number, shutter speed, ISO
 - Aperture-priority vs. shutter-priority
- AF: autofocus
 - Phase detection vs. contrast detection
- AWB: automatic white balance
 - How to identify which pixels are white?